I/A Practice 02

Title : It is to find out whether the length of the string for pendulum bob affects

$T = 2\pi\sqrt{\frac{l}{g}}$

Independent variable: Length of string

Dependable variable: Time

Precaution: 1. Small angle to prevent pendulum from wobbling

l/m ±0.01m	t1/ sec ± 0.01 s	t2 / sec +/− 0.01s	t3/ sec $\pm$ 0.01 s	t avg / sec $\pm$ 0.01 s	T / sec $\pm$ 0.01 s
0.10	6.69	6.67	6.60	6.66	0.67
0.20	9.04	9.19	8.94	9.06	0.91
0.30	11.10	11.09	11.06	11.08	1.11
0.40	12.69	12.38	12.53	12.54	1.25
0.50	13.91	13.91	13.85	13.89	1.39
0.60	15.81	15.84	15.72	15.79	1.58
0.70	17.06	16.94	17.44	17.15	1.71
0.80	18.12	18.32	18.03	18.16	1.82

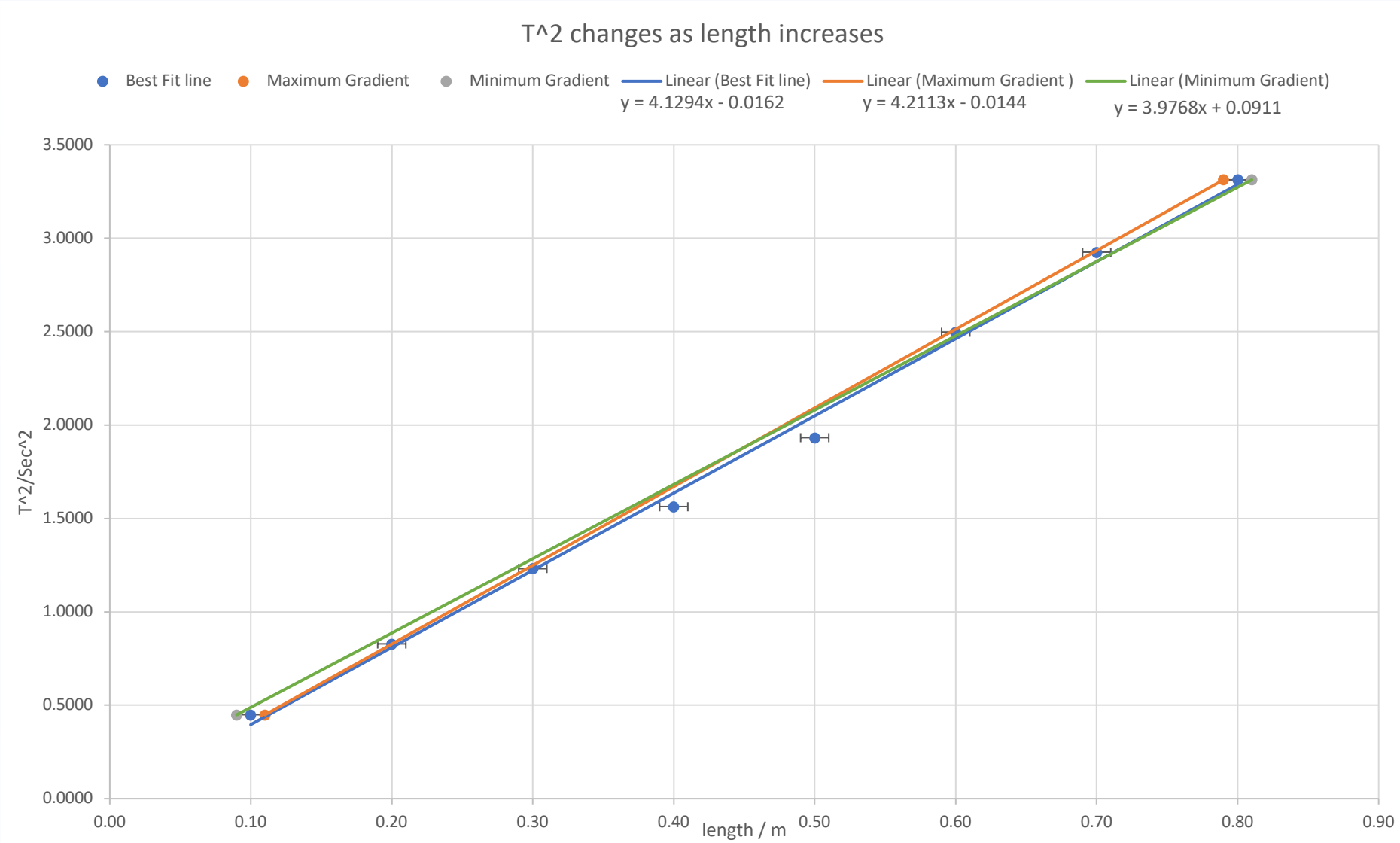
T^2 / sec^2	0.4489	0.8281	1.2321	1.5625	1.9321	2.4964	2.924	3.3124
l / m	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T^2 = 4\pi^2\frac{l}{g}$$

$$T^2 = \frac{4\pi^2}{g}\,l$$

Graph of T^2 against l



Data processing	Error propogation
$T^2 = \frac{4\pi^2(l)}{g}$ $g = \frac{4\pi^2(l)}{T^2}$ when l= 0.10 and T^2= 0.4489 $g = \frac{4\pi^2(0.10)}{0.4489}$ =8.7944 ≈8.79 When l = 0.20 and T^2= 0.8281 $g = \frac{4\pi^2(0.20)}{0.8281}$ =9.5346 ≈9.53 When l = 0.30 and T^2= 1.2321 $g = \frac{4\pi^2(0.30)}{1.2321}$ =9.6124 ≈9.61 When l = 0.40 and T^2= 1.5625 $g = \frac{4\pi^2(0.40)}{1.5625}$ = 10.1064 ≈ 10.11	$\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2} + \frac{\Delta l}{l}) \times 100\%$ $= (\frac{0.0001}{0.4489} + \frac{0.01}{0.10}) \times 100\%$ = 10.022% ≈ 10.0% $\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2} + \frac{\Delta l}{l}) \times 100\%$ $= (\frac{0.0001}{0.8281} + \frac{0.01}{0.20}) \times 100\%$ = 5.012% ≈5.01% $\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2} + \frac{\Delta l}{l}) \times$ $= (\frac{0.0001}{1.2321} + \frac{0.01}{0.30}) \times 100\%$ = 3.34144 ≈ 3.34% $\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2} + \frac{\Delta l}{l}) \times 100\%$

When l = 0.50 and T^2= 1.9321

g = $\frac{4\pi^2(0.50)}{1.9321}$

= 10.2164 $\approx$ 10.22

When l = 0.60 and T^2= 2.4964

g = $\frac{4\pi^2(0.60)}{2.4964}$

= 9.4884 $\approx$ 9. 49

When l = 0.70 and T^2= 2.9240

g = $\frac{4\pi^2(0.70)}{2.924}$

= 9.4510 $\approx$ 9.45

When l = 0.80 and T^2= 3.3124

g = $\frac{4\pi^2(0.80)}{3.3124}$

= 9.5346 $\approx$ 9.53

Average g value = $\frac{8.79+9.53+9.61+10.11+10.22+9.49+9.45+9.53}{8}$

= 9.5922

$\approx$ 9.59

= $(\frac{0.0001}{1.5625}+\frac{0.01}{0.40})\times 100\%$

= 2.506% $\approx$ 2.51%

$\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2}+\frac{\Delta l}{l})\times$

=$(\frac{0.0001}{1.9321}+\frac{0.01}{0.50})\times 100\%$

= 2.0051%$\approx$ 2.01%

$\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2}+\frac{\Delta l}{l})\times 100\%$

= $(\frac{0.0001}{2.4964}+\frac{0.01}{0.60})\times 100\%$

= 1.6706 %$\approx$ 1.67%

$\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2}+\frac{\Delta l}{l})\times 100\%$

=$(\frac{0.0001}{2.924}+\frac{0.01}{0.70})\times 100\%$

= 1.431% $\approx$ 1.43%

$\%\frac{\Delta g}{g} = (\frac{\Delta T^2}{T^2}+\frac{\Delta l}{l})\times 100\%$

=$(\frac{0.0001}{3.3124}+\frac{0.01}{0.80})\times 100\%$

= 1.55189% $\approx$ 1.55%

	Average percentage value of uncertainty $= \frac{1.55+1.43+2.01+1.67+2.51+3.34+5.01+10.0}{8}$ $= 3.44250 \approx 3.44\ \%$

$\therefore$ $value$ of uncertainty = $3.44\% \times 9.59$

$= 0.3301$

≈ 0.3

Actual g value = 9.81

% error = $\frac{9.81-9.59}{9.81} \times 100\%$

$= 2.2426$

$\approx 2.24\ \%$

$\therefore$ value for g = 9.59 ± 0.3